Field Hours

CSCE 672 Computer Supported Collaborative Work

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 90 minutes

Four cells, and the handoff

Locate the collaboration in time and in place before you pick a tool. The 4 cells are same time and place, same time and different places, different times and same place, and different times and places. A room board is the first. A live call is the second. A shift log on one ward is the third. Email is the fourth. A shared document edited hours apart serves the fourth cell and does not, by itself, carry a same-time argument.

Real work crosses cells, and the handoff is where systems fail. A same-time call can produce a decision a later shift must read. If nobody writes it into the shift log, cell 2 never reaches cell 3. You treat the handoff as an object with an author and a timestamp. A product that truly covers 1 cell should say so. Covering 1 cell is not a defect until you claim you covered 4.

Time and place, four cells.

Same time, same place

Same time, different places

Different times, same place

Different times, different places

A call that never reaches the shift

1. The decision is made on a live call. That is cell 2, same time and different places.

2. The next shift reads a log on the ward. That is cell 3. If the call is not written down, cell 3 never receives it.

3. You make the handoff an object with an author and a timestamp. A tool that only covers 1 cell must not claim it covers 4.

Problem 1. Which cell is a shared document that people edit hours apart, and why does it not by itself support a same-time argument?

Answer. Different times and different places, the fourth cell. Hours apart is not same time. You name the cell the tool actually serves. A same-time argument needs a different cell, such as a live call.

Problem 2. A decision is made in cell 2 and needed in cell 3. What object do you design, and when is covering only 1 cell a defect?

Answer. A handoff with an author and a timestamp, written into the shift log. If nobody writes it, cell 2 never reaches cell 3. Covering 1 cell is a defect only when you claim you covered 4.

Lecture 2 · 90 minutes

Awareness has a volume

Awareness is knowing who is here, what they did, and where they are in the shared thing. A feed of the last 5 edits answers whether someone changed the paragraph you are about to touch. You choose which events belong in those 5. A join, a leave, and an edit are different, and one undifferentiated list hides the edit. Design the feed as attention, not as a dump of every event the server can emit.

Too much awareness is a failure mode with a number. If 40 edits arrive in a minute, you cannot read them and also do the task. Aggregate, or slow the feed down to the events that touch your current section. Peripheral awareness, a marker that 3 people are in the document, is cheaper than a narration of their keystrokes. Keystroke narration is surveillance dressed as help, and it will change what people are willing to type.

A feed you can read, and one you cannot.

Last 5 edits

You chose which events

The edit stays visible

40 edits a minute

You cannot read and work

Aggregate or filter

Five events, not every key

1. The feed shows the last 5 edits that touch the section you have open. Joins are a separate marker.

2. A marker says 3 people are in the document. You do not stream their keystrokes.

3. At 40 edits a minute the 5-event rule still holds: you aggregate, you do not expand the feed until it becomes the task.

Problem 1. Why do you not put joins, leaves, and edits in one undifferentiated list of 5?

Answer. They are different events, and the list hides the edit. You choose which events belong in the 5. Awareness is a design of attention. The feed answers whether someone changed the paragraph you are about to touch.

Problem 2. 40 edits arrive in a minute, and a marker already says 3 people are present. What do you change, and why not narrate keystrokes?

Answer. Aggregate, or keep only events that touch the current section. You cannot read 40 edits and do the task. Keystroke narration is surveillance dressed as help, and it changes what people will type. The marker of 3 is the cheaper peripheral signal.

Lecture 3 · 90 minutes

Common ground is evidence

Common ground is a fact you and a partner mutually believe you share. A fact on the screen is not ground if the other person does not know you saw it. Evidence can be a phrase repeated back or a cursor both of you can see. At a table, a point can do that in 1 gesture. A remote pair may need 3 extra turns to confirm the same paragraph. You count those turns as task time, not as chatter.

A tool can cheapen grounding or counterfeit it. A green presence dot says someone is online. It is not evidence they see your paragraph. You have ground about a line only when there is evidence about that line. A shared highlight both people can see is stronger than 1 dot. If the product shows the dot and hides the selection, it bought the cheaper signal. Do not report the dot as common ground about a sentence.

Presence is not a shared referent.

One green dot

Says someone is online

Silent about the line

A shared highlight

Both see the sentence

Evidence about that line

Three turns to name a paragraph

1. A remote pair needs to confirm they mean paragraph 4. They spend 3 extra turns.

2. Those 3 turns are task time. A co-located pair might have pointed once, which is 1 gesture.

3. A green dot during the 3 turns does not replace them. Ground about paragraph 4 needs evidence about paragraph 4.

Problem 1. Why is a fact on the screen not yet common ground, and how many extra turns might a remote pair spend to confirm one paragraph?

Answer. The other person may not know you saw it. Common ground is mutual belief that you share the fact. A remote pair may need 3 extra turns, against 1 gesture at a table. You count those turns as task time.

Problem 2. A product shows 1 green dot and hides the selection. What have you not established?

Answer. Common ground about a sentence. The dot says someone is online. Ground about a line needs evidence about that line, such as a highlight both people can see. The product bought the cheaper signal.

Lecture 4 · 90 minutes

Articulation is the work around the work

Articulation work divides, sequences, and repairs a shared task. It is not the task itself. Someone must decide who writes the introduction, what done means, and what happens after a silence of 2 days. A checklist of 6 steps makes that work visible. If the software hides the checklist, the work moves into memory and private notes. You

will see missed steps. The list was coordination, even when it looked like extra process.

A handoff packages state for the next person when the time-place cell changes. A ticket with 3 required fields, owner, status, and next action, is a small version of that package. If next action is optional, tickets sit in progress with no verb. You require the field. Automatic assignment among 4 people is fine only when the rule is readable, for example a round-robin. A hidden assignment removes a negotiation people were actually doing.

A ticket that carries the handoff.

Name the owner -> Set the status -> Require the next action -> Show the assignment rule

Six steps that were not bureaucracy

1. The group uses a checklist of 6 steps to decide who writes next after a silence of 2 days.

2. The new tool hides the checklist and opens the document directly. Missed steps show up within the week.

3. You put the checklist back, or you encode its decisions as required ticket fields. Hiding it did not delete the articulation.

Problem 1. A checklist has 6 steps and the software hides it. Where does the articulation go, and what do you observe?

Answer. It moves into memory and private notes. You will see missed steps. The list was coordination, not extra process, including the decision about a silence of 2 days.

Problem 2. Next action is optional on a ticket, and assignment among 4 people is hidden. What two failures are those?

Answer. Tickets sit in progress with no verb, so you require the next-action field. Hidden assignment removes a negotiation. A readable rule, such as round-robin among the 4, keeps the articulation visible.

Lecture 5 · 90 minutes

Two inserts, one string

You say what two simultaneous writes mean. Last-write-wins on the whole string is easy and lossy. Both start from 'abcd', length 4. One inserts X at index 2, producing 'abXcd'. The other inserts Y at index 2, producing 'abYcd'. Keeping only the later full string leaves length 5, not 6. The author of the lost insert may get no notice, so this is an awareness failure as well as a consistency failure.

You keep both inserts by transforming indexes under a shared tie rule. If both target index 2, the loser of the tie shifts by 1. Both replicas can then build one string of length 6, either 'abXYcd' or 'abYXcd'. If the inserts start at different indexes, 1 and 3, the greater index grows by the earlier insert's length. A one-character insert ordered first at index 1 turns the other index from 3 into 4.

Same two inserts, two storage rules.

Last write of the string

Keeps one full string

Length becomes 5

One character is lost

Index transform

Shifts the losing index by 1

Length becomes 6

Both characters remain

Index 3 becomes 4

1. The string 'abcd' has length 4. One insert is a single character at index 1. The other is a single character at index 3.

2. The tie rule orders the index-1 insert first. The other index grows by 1, from 3 to 4.

3. Both characters remain, so the length is 6. Last-write-wins on the whole string would have left length 5.

Problem 1. Both people insert one character at index 2 into 'abcd'. What lengths do last-write-wins and an index transform produce?

Answer. Last-write-wins keeps one full string, length 5, and drops a character. The transform shifts the losing index by 1 and keeps both characters, length 6, as 'abXYcd' or 'abYXcd' depending on the tie.

Problem 2. Inserts are one character at index 1 and one character at index 3. The index-1 insert is ordered first. What does the other index become, and what is the final length?

Answer. The index 3 becomes 4, because it grows by the length of the earlier insert, which is 1. Both characters remain, so the length is 6, starting from 4.

Lecture 6 · 90 minutes

Skew, visibility, and consent

Count the social pattern, do not only describe it. On a document with 8 accounts, 2 of them may make 90 percent of the edits. The other 6 are not automatically a failure. A newcomer who reads for 7 days and then edits is not the same as an owner who never comes back. You need the sequence, not only the totals 2 and 6. One snapshot cannot separate learning from free riding.

Visibility changes contribution and it spends privacy. Publishing each edit count can raise output and can punish someone who revises carefully. A marker that shows you in the file at midnight is a different disclosure from a cursor in a scheduled hour. You need consent for the stronger one. Awareness that helps coordination is not automatically acceptable. If the 8 people did not agree to overnight presence, an account signup was not that agreement.

A cursor at work, and a dot at midnight.

Scheduled cursor

Shows the shared hour

Matches the task

Midnight marker

Shows presence at night

Needs its own consent

Two editors out of eight

1. The document has 8 accounts. Across the whole history, 2 accounts made 90 percent of the edits.

2. The weekly sequence shows one of the other 6 editing after 7 days of reading. That person is not the same pattern as an owner who never returns.

3. You report 2 and 6 and the sequence. You do not invent a motive the log does not show.

Problem 1. 2 of 8 accounts make 90 percent of the edits. Why is the other 6 not automatically free riding?

Answer. A snapshot of 2 and 6 cannot separate stories. A newcomer who reads for 7 days and then edits is not an owner who never comes back. You need the sequence before you name a motive.

Problem 2. Why is a midnight presence marker not justified by the fact that awareness helps coordination, if 8 people only signed up for an account?

Answer. It is a stronger disclosure than a cursor in a scheduled hour. Awareness that helps coordination is not automatically acceptable. Signup was not consent to overnight presence. You need consent for the stronger sensor.

Lecture 7 · 90 minutes

Study the group before you automate

Study the group before you automate the coordination. An interview with 1 manager yields the official plan, which is often not the plan in use at 2 in the morning. Keep a diary of 5 days of handoffs. You will see steps a chart skips, such as a call that carries the real next action while the ticket stays empty. If you automate the empty ticket, you freeze the wrong process.

The unit you study is the group. A test of 1 person on 1 editor can be careful and still miss the handoff between 2 roles. Give a pair a task that forces that handoff. Record where they build common ground and where they fail. 3 extra turns spent agreeing on a referent belong in the findings. A note that only says a button was hard to see has stayed inside a single-user analysis.

Watch the handoff, then automate.

1. Diary of 5 days

2. Note the call the ticket missed

3. Test a pair, not 1 user

4. Count turns spent grounding

The call that was the process

1. For 5 days the real next action travels by a phone call. The ticket stays empty.

2. An interview with 1 manager describes the official 6-field ticket and never mentions the call.

3. If you automate the empty ticket, you freeze the wrong process and drop the call. A pair test that records 3 grounding turns will see the handoff the single-user button test missed.

Problem 1. You interviewed 1 manager and the ticket is empty while a call carries the next action. What happens if you automate that ticket?

Answer. You freeze the wrong process and drop the call that made the work succeed. The manager gave the official plan, often not the plan in use at 2 in the morning. A diary of 5 days is what shows the call.

Problem 2. A careful test of 1 person finds a hard button, and a pair spends 3 turns agreeing on a referent. Which finding belongs to this course, and why is the button note not enough?

Answer. The 3 turns belong in the findings, because the unit is the group and the handoff between 2 roles. A note that only says the button was hard stays inside a single-user analysis. It never asks who else had to understand the work.

Lecture 8 · 90 minutes

A workflow with an explicit limit

You can now place work in the time-place matrix and name the cell a tool really serves. You can size an awareness feed, for example the last 5 events, and reject a feed of 40 events a minute. You can treat a checklist as articulation and keep 2 simultaneous inserts without dropping one. You can report a skew of 2 editors out of 8 without inventing a motive. You can refuse to automate a handoff you have not watched.

The cell and the write rule interact. Different-time editing makes overlapping inserts likely, because the authors do not share a cursor. Last-write-wins then keeps length 5 and drops a character that an index transform would have kept to reach length 6. You either add same-time awareness so they see the clash, or you keep both characters. A feed of 5 events does not repair a rule that deletes one of them.

Name the limit of the workflow.

Name the time-place cell -> Size the awareness feed -> Pick a write rule -> Watch before you automate

Async edits and a dropped character

1. The document is different-time and different-place. The authors do not share a cursor, so overlapping inserts are likely.

2. Both insert into 'abcd'. Last-write-wins leaves length 5. A transform leaves length 6.

3. You either move the activity toward same-time awareness or you keep both characters. Showing the last 5 events does not restore the dropped character.

Problem 1. Two authors edit at different times, with no shared cursor, and both insert one character

into 'abcd'. What length does last-write-wins leave, what length does an index transform leave, and why does a feed of 5 events not fix the shorter result?

Answer. Last-write-wins leaves length 5 and drops a character. The transform leaves length 6. Different-time work makes the overlap likely because they do not see each other's cursor. A feed of 5 events is awareness, not a storage rule, so it does not put the dropped character back. You either add same-time awareness or you keep both characters.

Problem 2. A log shows 2 of 8 accounts editing, and you have not watched the handoff. What may you report, and what do you refuse to automate?

Answer. You may report the skew, 2 out of 8, without inventing a motive the log does not show. You refuse to automate the handoff until you have watched it. The checklist and the diary are how you learn the articulation. The transform, if you use one, must still be correct even when only 2 people edit.